



Lab Task – 4

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Course: Programming Fundamentals – Lab

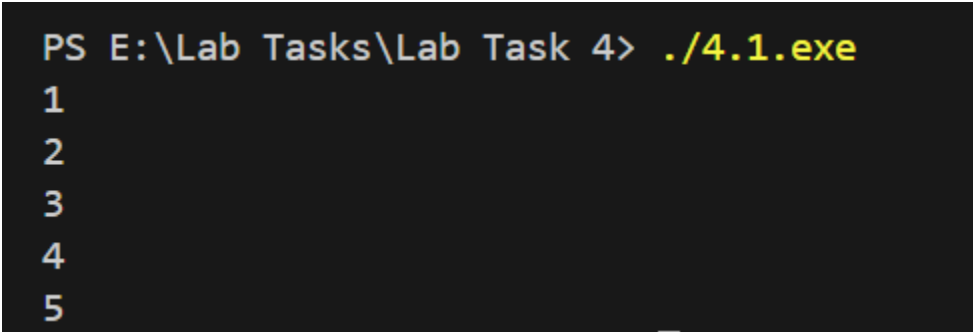
Course Instructor: Ms. Sadaf Ayub

Task – 1

```
#include<iostream>
using namespace std;
int main()
{
    for ( int i = 1; i <= 5; i++ )
    {
        cout << i << endl;
    }

    return 0;
}
```

Output



```
PS E:\Lab Tasks\Lab Task 4> ./4.1.exe
1
2
3
4
5
```

Task – 2

```
#include<iostream>
using namespace std;
int main()
{
    int n;
    int sum = 0;
    cout << "Enter a positive integer: ";
    cin >> n;

    for ( int i = 1; i <= n; i++ )
    {
        sum = sum + i;
    }

    cout << "The sum of first " << n << " natural
numbers is " << sum << endl;

    return 0;
}
```

Output

```
PS E:\Lab Tasks\Lab Task 4> ./4.2.exe
Enter a positive integer: 10
The sum of first 10 natural numbers is 55
```

Task – 3

```
#include<iostream>
using namespace std;
int main()
{
    int num;
    cout << "Enter the number you want the
multiplication table of: ";
    cin >> num;

    for ( int i = 1; i <= 10; i++ )
    {
        cout << num << " x " << i << " = " << num * i
<< endl;
    }

    return 0;
}
```

Output

```
PS E:\Lab Tasks\Lab Task 4> ./4.3.exe
Enter the number you want the multiplication table of: 9
9 x 1 = 9
9 x 2 = 18
9 x 3 = 27
9 x 4 = 36
9 x 5 = 45
9 x 6 = 54
9 x 7 = 63
9 x 8 = 72
9 x 9 = 81
9 x 10 = 90
```

Task – 4

```
#include<iostream>
using namespace std;
int main()
{
    for ( int i = 1; i < 20; i++ )
    {
        if ( i % 2 == 0 )
        {
            cout << i << " ";
        }
    }
}
```

```
    return 0;
}
```

Output

```
PS E:\Lab Tasks\Lab Task 4> ./4.4.exe
2 4 6 8 10 12 14 16 18
```

Task – 5

```
#include<iostream>
using namespace std;
int main()
{
    for ( int i = 10; i >= 1; i--)
    {
        cout << i << " ";
    }

    return 0;
}
```

Output

```
PS E:\Lab Tasks\Lab Task 4> ./4.5.exe  
10 9 8 7 6 5 4 3 2 1
```

Task – 6

```
#include<iostream>  
using namespace std;  
int main()  
{  
    int num;  
    long long fact = 1;  
    cout << "Enter the positive number you want to  
find the factorial of: ";  
    cin >> num;  
  
    if ( num < 0 )  
    {  
        cout << "Factorial of negative numbers does  
not exist.";  
    }  
    else  
    {  
        for ( int i = 1; i <= num; i++ )  
        {
```

```
        fact = fact * i;
    }

    cout << "Factorial of " << num << " is " << fact
<< endl;
}

return 0;
}
```

Output

```
PS E:\Lab Tasks\Lab Task 4> ./4.6.exe
Enter the positive number you want to find the factorial of: -5
Factorial of negative numbers does not exist.
PS E:\Lab Tasks\Lab Task 4> ./4.6.exe
Enter the positive number you want to find the factorial of: 7
Factorial of 7 is 5040
```

THE END

